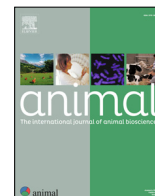




Animal

The international journal of animal biosciences



Multifaceted analysis of temporal and spatial distribution and risk factors of global poultry HPAI-H5N1, 2005–2023



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ARTICLE INFO

Article history:

Received 22 October 2023

Revised 15 January 2024

Accepted 15 January 2024

Available online 24 January 2024

Keywords:

HPAI-H5N1

Maximum entropy model

Risk factors

Time series decomposition

Spatial distribution

ABSTRACT

The purpose of this study was to analyze the characteristics of occurrence and spread of highly pathogenic avian influenza H5N1 (**HPAI-H5N1**) globally, understand its spatiotemporal characteristics, investigate the risk factors influencing outbreaks, and identify high-risk areas for disease occurrence. We collected the data on global poultry HPAI-H5N1 outbreaks from January 2005 to April 2023, and conducted a thorough analysis of the spatial and temporal characteristics of the disease through time series decomposition and directional distribution analysis. Additionally, an ecological niche model was established to explore the major factors influencing the occurrence of HPAI-H5N1 and to pinpoint high-risk areas. Our findings revealed that HPAI-H5N1 outbreaks were cyclical, and seasonal, exhibiting a rising trend, with a predominant northwest-southeast transmission direction. The ecological niche model highlighted that species factors and economic trade factors are critical in influencing the outbreak of HPAI-H5N1. Variables such as chicken and duck density, population density, isothermality, and road density, contributed to importantly risk of outbreaks. High-risk areas for HPAI-H5N1 occurrence were primarily identified in Europe, West Africa, Southeast Asia, and Southeast China. This study provided valuable insights into the spatial and temporal distribution characteristics and risk factors of global poultry HPAI-H5N1 outbreaks. The identification of high-risk areas provides essential information that can be used to develop more effective prevention and control policies.

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Implications

Currently, Highly Pathogenic Avian Influenza H5N1 is widely prevalent internationally and has caused outbreaks in a large area, resulting in important economic losses in the poultry industry. At the same time, as a zoonotic disease, it constantly poses a threat to human health. This study establishes an ecological niche model for Highly Pathogenic Avian Influenza H5N1 virus and investigates the social, economic, trade, environmental, and geographical factors that affect the outbreak of the disease. It predicts the suitable geographical range of the virus, providing a theoretical basis and guidance for Highly Pathogenic Avian Influenza H5N1 prevention and control in various countries.

Introduction

Highly Pathogenic Avian Influenza H5N1 (**HPAI-H5N1**) was first isolated from domestic geese in Guangdong, China, in 1996, mark-

ing the first identification of this particular subtype (Bui et al., 2018; Vijaykrishna et al., 2008). The virus subsequently caused its first major outbreak in Hong Kong, China, in 1997. Between 2004 and 2006, avian influenza (**AI**) rapidly spread across Asia and eventually led to a widespread global pandemic (Peiris et al., 2007). In response to this threat, countries worldwide implemented various measures to prevent and control HPAI-H5N1, yet they continue to face periodic recurrences of the disease (Gilbert et al., 2008). Recently, there has been another wave of HPAI-H5N1 outbreaks globally, with frequent incidents reported in Europe (Haider et al., 2023; Nagy et al., 2023), the Americas (Ruiz-Saenz et al., 2023), and the West African region (FAO, 2022).

The prevalence of avian influenza virus (**AIV**) in domestic poultry has considerably burdened global economies while posing a threat to the health of humans, livestock, and wildlife (Gilbert and Pfeiffer, 2012; Gilbert et al., 2014; Vergne et al., 2014). HPAI-H5N1 is a zoonotic infection with a mortality rate of up to 100% in poultry (Chen et al., 2022). Typically, AIV is transmitted among birds and poultry through droplets and direct contact (Lycett et al., 2019). The viral ability to survive in water for extended periods increases the risk of infection in mammals,

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including humans. However, mammal-to-mammal and human-to-human transmission remains rare (Alexander, 2007).

Both domestic poultry and wild birds have played pivotal roles in the global dissemination of HPAI-H5N1 (Peterson and Williams, 2008). Among wild birds, those belonging to the Anseriformes order (including ducks, geese, and swans) and Charadriiformes category (comprising gulls and shorebirds) have been identified as primary carriers of AIV (Jennelle et al., 2016). The emergence of highly pathogenic avian influenza (HPAI) in poultry is widely attributed to the initial presence of Low Pathogenic Avian Influenza (LPAI) in wild migratory birds, which subsequently mutated into HPAI upon introduction to poultry populations (Alexander, 2007; Grillo et al., 2015). During migration, wild birds often come into contact with domestic poultry, facilitating the transfer of AIVs into poultry populations (Xiao et al., 2007). The adaptation of LPAI to the diverse wild bird populations is attributed to differences in host density and environmental conditions. In contrast, LPAI is more prone to evolve into HPAI in artificially raised poultry populations due to distinct ecological factors (Lebarbenchon et al., 2010).

The occurrence of HPAI results from a complex interplay of various factors, including human and vehicle movements, just migratory birds, and the farm breeding environment (Han et al., 2021). The migration of wild birds plays a critical role in dispersing and preserving viruses in the wild (Gauthier-Clerc et al., 2007). Ferenczi et al., (2021) demonstrated that increased rainfall intensity in Australia can lead to the spillover of Avian Influenza Virus (AIV) from wild birds to poultry, thus heightening the risk of AI infection in poultry. Radin et al., (2017) observed that the global trade of live chickens escalates the risk of HPAI outbreaks in importing countries. Human factors, particularly in densely populated areas and large cities, are greatly influential in the spread of H5N1 (Paul et al., 2010). Gilbert et al. found that in Southeast Asia, the risk of HPAI-H5N1 is linked with duck density, population density, and rice cultivation intensity, whereas its correlation with chicken density is comparatively weak (Gilbert et al., 2008). In Thailand, a study identified the density of fighting cocks, quail, free-ranging duck flocks, and poultry slaughterhouses as key risk factors influencing HPAI-H5N1 outbreaks (Tiensin et al., 2009). Similarly, in China, Martin et al. discovered that HPAI-H5N1 outbreaks in poultry are primarily related to chicken density, population density, and altitude (Martin et al., 2011). However, the pathways and significance of these factors at a global scale remain ambiguous (Kilpatrick et al., 2006). Therefore, it is crucial to investigate the key factors affecting the occurrence and spread of HPAI-H5N1 on a large geographic scale.

Time series data analysis is an essential tool for understanding the dynamics of infectious diseases, though it has limitations in predicting disease outbreaks (Brown et al., 2009; Trawinski and Mackay, 2008). This analysis method involves the decomposition of a dataset to identify time-scale dependent features that influence the observed occurrence of a disease or event. Decomposition separates these features for a more detailed analysis. One key approach in time series analysis is using autoregressive models. These models assess how much of the current trends in the data can be explained by previous observations within the same time series. The patterns identified in these models typically encompass long-term trends, cyclical movements, and seasonal variations. Understanding these temporal patterns, including trends and seasonality, is crucial for grasping the underlying causes of a disease's spread or emergence (Ward et al., 2020).

In this study, we established a monthly time series database of global poultry HPAI-H5N1 events from January 2005 to April 2023. We employed regression models in our time series analysis to characterize the occurrence data of HPAI-H5N1, focusing on identifying seasonality and trends in the disease's prevalence. A crucial

part of our analysis involved examining the spatial distribution of HPAI-H5N1 incidents in each identified period, aiming to elucidate the spatial characteristics and transmission directions of HPAI-H5N1 in each cycle. Subsequently, we engaged in modeling the maximum entropy ecological niche. This model helped us explore the important factors that influence the occurrence and spread of poultry HPAI-H5N1 globally. We investigated these factors across four dimensions: economy and trade, species distribution, geographical landscape, and bioclimate. Our analysis also included predicting potential HPAI-H5N1 suitability distribution areas. The ultimate goal of this study is to provide a robust theoretical foundation for the surveillance, prevention, and control of global poultry HPAI-H5N1.

Material and methods

Data

Data on the HPAI-H5N1 outbreak were obtained from the World Organization for Animal Health (WOAH) and the Food and Agriculture Organization of the United Nations (FAO). The outbreak report included crucial information such as the country and region of the outbreak, the geographical coordinates (latitude and longitude) of the disease occurrence, the time of the outbreak, and the types of animals infected. The data collection spanned from January 2005 to April 2023. Reports on wildlife and mammalian disease occurrences were excluded, focusing solely on poultry disease records, resulting in a total of 3 045 HPAI-H5N1 disease reports pertaining to poultry (Supplementary Figure S1).

Time series decomposition

Time series models are valuable for analyzing, describing, and predicting diseases. In this study, we utilized R 4.3.0 to write the code for time series decomposition. This enabled us to integrate data and establish a comprehensive monthly time series dataset of global poultry HPAI-H5N1 occurrences from January 2005 to April 2023. The dataset comprised 220 monthly data points, each indicating the number of incidence events for that month.

For the decomposition of the time series of global poultry H5N1-HPAI incidence events, we employed the Seasonal and Trend decomposition using Loess (STL) method. STL is a model based on the additive principle, expressed as $Y_{[t]} = T_{[t]} + S_{[t]} + R_{[t]}$. In this formula, $Y_{[t]}$ represents the model output at time t , and $T_{[t]}$, $S_{[t]}$, and $R_{[t]}$ denote the trend component, the seasonal component, and the residuals at time t , respectively.

The R packages for data visualization, manipulation and analysis of time-series data used in this analysis: ggplot2 (Wickham, 2016), plyr (Wickham, 2011), dplyr (Wickham, et al., 2019), lubridate (Grolemund and Wickham, 2011), tseries (Trapletti and Hornik, 2018).

Directional distribution analysis

To analyze the directional distribution of historical HPAI-H5N1 epidemics and to determine their spread direction, we utilized SD ellipses. This analytical technique involves generating an elliptical surface by calculating the standard distances from a collection of case sites. By examining the major and minor axes, along with the direction of the ellipse, we can determine if the distribution of HPAI-H5N1 epidemic spots exhibits a directional trend. According to the results of Chen et al., global H5N1 has the characteristics of repeated outbreaks about every 5 years. Concurrently, the results of time series decomposition also confirm a five-year cycle in the outbreaks of global H5N1 influenza. Hence, the global epi-

demioleological data of HPAI-H5N1 was segmented into four distinct periods for a detailed analysis: 2005–2009, 2010–2014, 2015–2019, and 2020–2023. This segmentation facilitated a phase-wise examination of the disease's spread. Each SD ellipse, representing a respective phase, encapsulates 95% of the data points within its cluster. For further insights into directional distribution analysis, consult [Supplementary Material S1](#).

Collection of model variables

To establish an ecological niche model for analyzing the key factors affecting disease occurrence, we selected a total of 26 variables, categorized into four types: bioclimatic factors, economic and trade factors, species factors, and geographic landscape factors.

Bioclimatic factors were sourced from 19 variables in the WorldClim 2.1 database. We assessed multicollinearity and Pearson correlation coefficients of these variables using the Uncertainty Analysis for Species Distribution Models package in R 4.3.0. Variables with correlation coefficients exceeding 0.7 and variance inflation factor (VIF) greater than 5 were excluded, leaving us with 8 bioclimatic variables. The VIF is a measure of the severity of complex (multiple) covariance in a multiple linear regression model. It represents the ratio of the variance of the estimated regression coefficients to the variance if the independent variables are assumed not to be linearly correlated.

Economic and trade factors comprised road and railway data from the Resource and Environment Science and Data Center, as well as global population density data available at [<https://landscan.ornl.gov/>]. Kernel density analysis was conducted on the road and railway vector data, setting a search radius of 10 km to generate road density and railway density raster data. Species factors included global chicken and duck density, obtained from [<https://www.fao.org/livestock-systems/en/>]. Geographic landscape factors were derived from the WorldClim 2.1 database for elevation and from [<https://livingatlas.arcgis.com/landcoverexplorer/>] for global land cover type ([Supplementary Table S1](#)). A total of 15 variables were finalized for inclusion in the ecological niche model ([Supplementary Table S2](#)). These 15 raster variables, sourced from various databases, were standardized using ArcGIS 10.2 software. We ensured uniformity by cropping all variables to the same extent, adjusting them to a uniform pixel size (5 arcmin resolution), and aligning them to a consistent coordinate system using the resampling tool.

Ecological niche modeling

To build the ecological niche model, it was necessary to reduce the occurrence of spatial clustering. To achieve this, spatial points were filtered using ENMTools ([Warren et al., 2010](#)) with a range of 10 kilometers. This process resulted in retaining 1 725 case reports, which were then included in the ecological niche model. S2 (An introduction to the maximum entropy model).

A maximal entropy ecological niche model was developed using MaxEnt.3.4.1 software, to analyze factors influencing the occurrence and spread of global poultry HPAI-H5N1 and forecast disease suitability distribution areas. The filtered disease occurrence data were divided into two parts, 80% as training set and 20% as test set. To prevent over-fitting of the model, the regularization constant was set to 1. Replicated run type was set to bootstrap. The model was run 100 times to reduce the error, and the average of the 100 times results was taken as the final prediction result. The subject operating characteristic curve (Area Under Curve-Receiver Operating Characteristic Curve (AUC-ROC)) was used as the final model performance evaluation metric. The significance of each variable was assessed using the percentage contribution

of the variables. ArcGIS.10.2 software was used for final visualization.

Result

Time series analysis

We compiled a time series database of 3 045 case events of HPAI-H5N1, spanning from January 2005 to April 2023. We plotted the HPAI-H5N1 monthly event time series dataset and superimposed a smooth curve to visually assess the trend. [Fig. 1](#) in our study illustrated both the HPAI-H5N1 occurrence events and the smoothing curve. The results indicated an overall increasing trend in the disease's occurrence throughout the study period.

The time series decomposition results revealed a periodicity in the outbreaks of HPAI-H5N1, with a cycle of approximately five years ([Fig. 2](#)). Globally, the data showed an increasing trend in the latest cycles of the disease, suggesting a potential shift in the dynamics of HPAI-H5N1 occurrences. Moreover, our analysis identified a clear seasonal pattern in the occurrence of HPAI-H5N1 in global poultry. The disease occurred most frequently in winter and spring, with occurrences being less frequent in summer and fall ([Fig. 3](#)).

Directional distribution analysis

During the first phase of our study (2005–2009), we observed a total of 319 poultry AI outbreaks. These outbreaks were predominantly concentrated in southern China, Southeast Asia, the Middle East, and southeastern West Africa. The epidemic's spread during this period followed a northwest-southeast direction, as illustrated in [Fig. 4](#) and detailed in [Table 1](#). In the second phase (2010–2014), the frequency of outbreaks increased to 413. The primary areas affected during this stage were the Malaysian archipelago, the central coastal region of China, the Korean Peninsula, the Japanese islands, and Egypt. Similar to the first phase, the epidemic's distribution pattern continued to be in a northwest-southeast direction. The third phase (2015–2019) marked an important rise in the number of outbreaks, with a total of 1 159 occurrences. The geographical spread of these outbreaks was mainly in Southeast Asia, Egypt, and Southeast West Africa. During this period, the epidemic maintained its northwest-southeast directional distribution. In the fourth and most recent phase (2020–2023), there were 1 154 poultry AI outbreaks. This phase saw a shift in the primary regions affected, with notable occurrences in the Korean Peninsula, the Japanese islands, southeastern West Africa, the United States, Mexico, and Europe. A key observation in this phase was the change in the epidemic's distribution pattern, which shifted to a roughly west-east direction.

Maximum entropy model evaluation and effect of chosen variables

The training set average AUROC value was 0.953 and the test set average AUROC value was 0.916, indicating the model's strong ability to distinguish between areas of global poultry HPAI-H5N1 ([Supplementary Figure S2](#)).

As outlined in [Table 2](#), 15 variables were ultimately selected for inclusion in the maximum entropy model. The importance of these variables was quantified using their contribution rates. Species variables accounted for the highest total contribution rate at 66.1%, followed by economic and trade variables at 20.5%, bioclimatic variables at 10.7%, and geographical landscape variables at 2.6%.

Among these, the variable with the highest individual contribution rate was chicken density (47.6%), followed by duck density

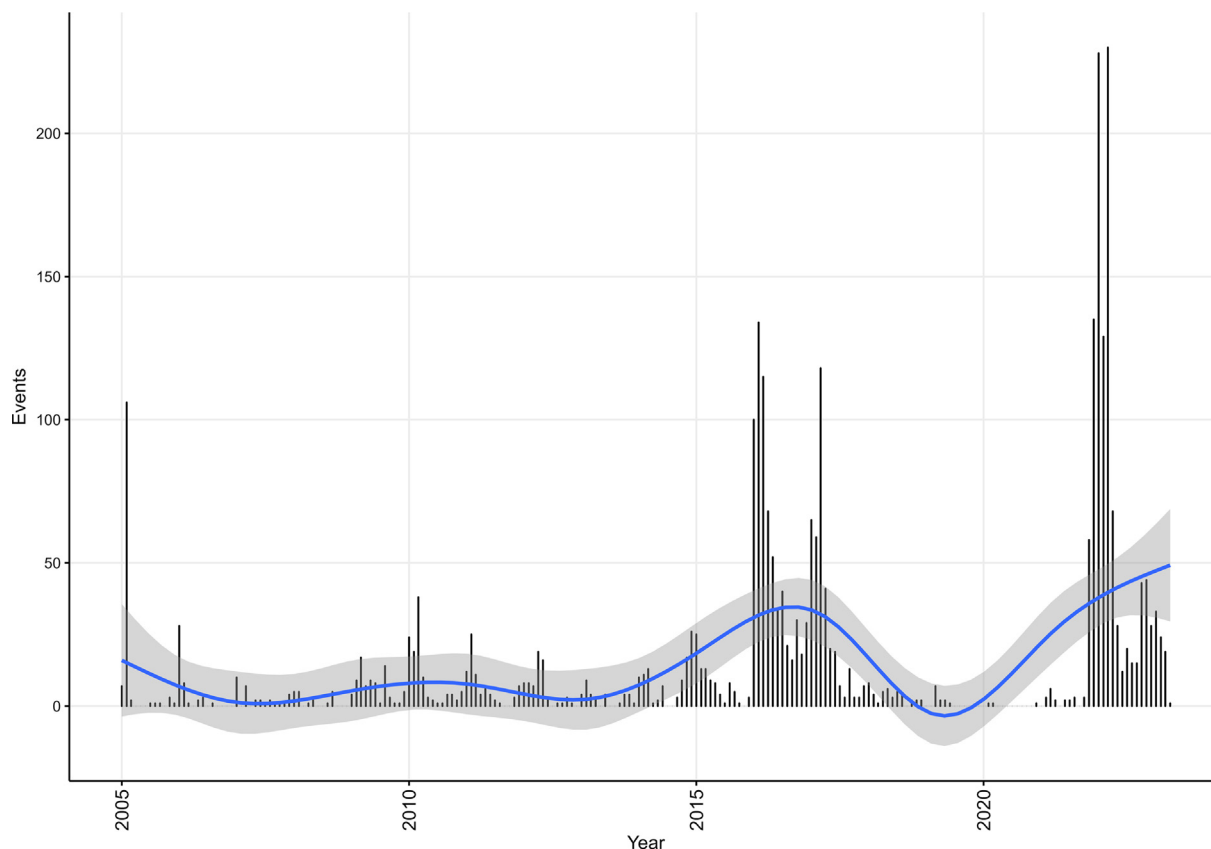


Fig. 1. Global poultry HPAI-H5N1 monthly incidence events from 2005 to 2023. The blue line represents a smooth curve with a 95% confidence interval (gray).

(18.5%), population density (16.9%), isothermality (4.7%), and road density (2.3%). These five variables alone, with a combined total contribution rate of 90%, are considered critical factors influencing the occurrence and spread of global poultry HPAI-H5N1.

We classified the suitability areas as high-risk and the unsuitability areas as low-risk. A binary method was employed to indicate the risk level, where 0 represents no risk and 1 represents the highest risk. According to our analysis, high-risk areas for HPAI-H5N1 outbreaks include Southeast China, southern and coastal Vietnam, Cambodia, Thailand, Bangladesh, eastern India, the Malay Peninsula, the Indonesian archipelago, Jordan, southeastern West Africa, France, the United Kingdom, and Germany. Moderate risk levels were identified in eastern and northern China, Japan, Korea, and eastern and central Europe, as well as parts of the Americas (Fig. 5).

Discussion

HPAI-H5N1 is a severe zoonotic disease with a high mortality rate, posing great threats to human and animal health, particularly affecting poultry and wild migratory birds (Krammer and Schultz-Cherry, 2023).

Using the STL method (Chen et al., 2022), we analyzed HPAI-H5N1 and found that outbreaks of this disease occur in approximately five-year cycles. Seasonal analysis results indicated that global poultry HPAI-H5N1 exhibits obviously seasonality, with peak occurrences from November to April of the following year, aligning with winter and spring seasons.

This seasonal variation is likely linked to the migratory patterns of wild birds (FAO, 2022; Normile, 2006; Xiao et al., 2007). In regions north of 25 ° N latitude, declining temperatures have been observed to accelerate the spread of AIV (Liu et al., 2007). Studies

have demonstrated that HPAI-H5N1 outbreaks tend to peak in environments with temperatures below 20 °C, following seasonal patterns (Nuñez and Ross, 2019).

In our study, the evolution of HPAI-H5N1 was segmented into four phases for a detailed directional distribution analysis. We observed that the epidemic intensity in the first two phases was lower compared to the latter two phases. This pattern indicated an increasing trend in the incidence of HPAI-H5N1 in global poultry, consistent with the trend represented by the smooth curve in Fig. 1. Directional analysis results revealed that the transmission direction of the epidemic in all four stages predominantly followed a northwest-southeast distribution. Since 2003, HPAI-H5N1 has expanded from East Asia to broader Asia and Europe, reaching as far as the United Kingdom, West Africa, and further spreading to the Americas (Bevins et al., 2022). This expansive spread is likely linked to the southwest migration patterns of migratory birds (Meng et al., 2019; Caliendo et al., 2022; Hill et al., 2022). The detection of H5N1 in Senegal is indicative of AIV spreading from Eurasia to West Africa (Lo et al., 2022), and in Africa, there is evidence of the westward spread of the virus (FAO, 2021). Therefore, the directional distribution analysis conducted in our study accurately reflects the transmission direction of HPAI-H5N1.

Prior research on the risk factors associated with poultry HPAI-H5N1 outbreaks has predominantly concentrated on specific countries or regions. Nonetheless, given the disease's potential for inter-continental transmission (Bevins et al., 2022), it is imperative to approach the study of these risk factors from a global perspective. In light of this, our study aimed to identify and investigate the key causes influencing the outbreak of poultry HPAI-H5N1 on a world-wide scale.

The findings from our ecological niche model shed light on the important factors contributing to the occurrence of poultry HPAI-

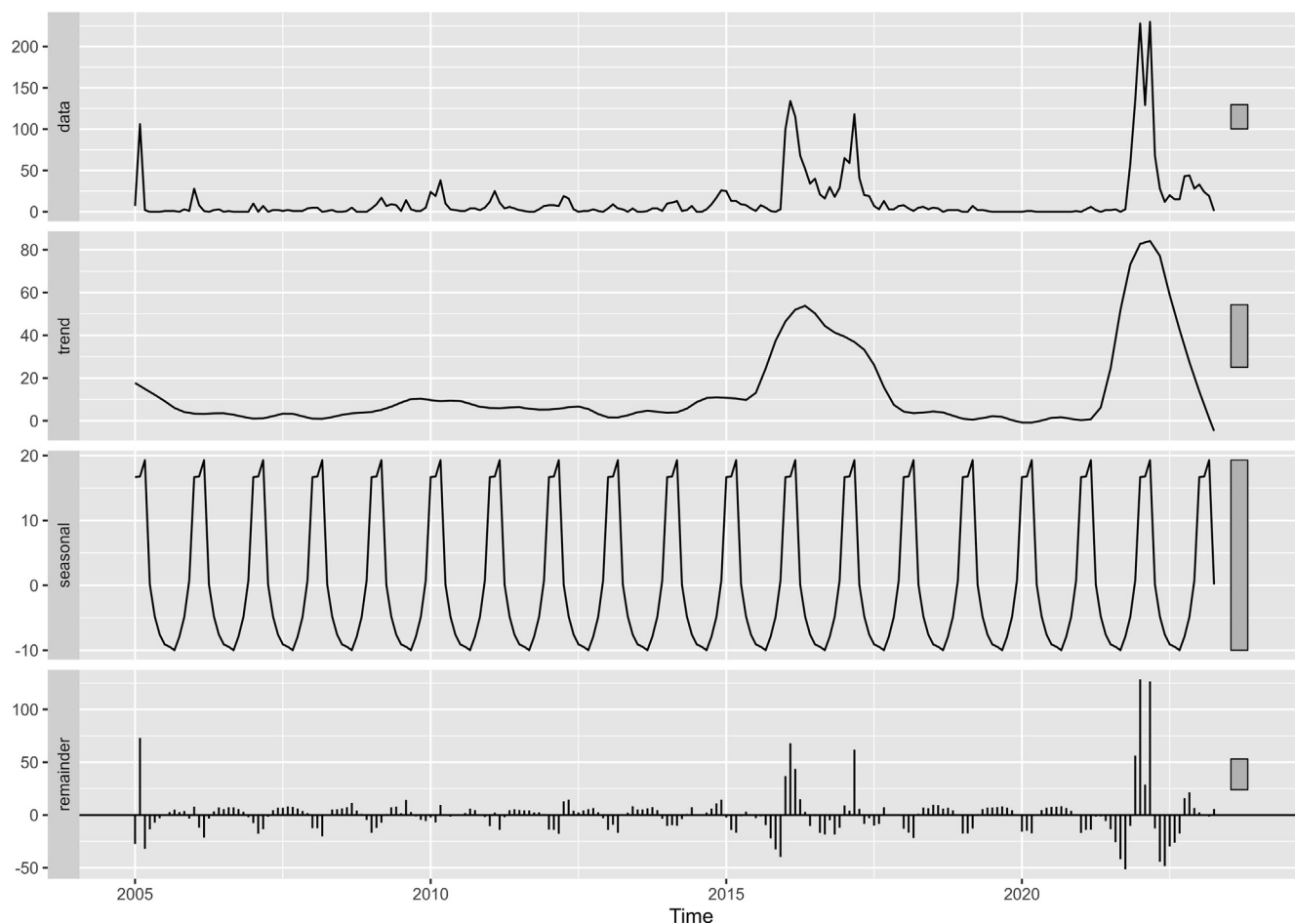


Fig. 2. Time series components of global poultry HPAI-H5N1 events from 2005 to 2023. X-axis represents time. The Y-axis represents the disease event, trend, seasonality, and residual components, respectively.

H5N1 globally. Species variability emerged as the most influential factor, accounting for 66.1% of the total impact. Economic and trade variables also played a substantial role, contributing 20.5% to the risk of outbreaks. In comparison, bioclimatic and geographic landscape factors had a relatively minor influence, together constituting less than 15% of the total contribution. These insights underscore the importance of species diversity and economic and trade activities in driving the global spread of HPAI-H5N1 in poultry. They also suggest that while climate and geographic factors do play a role, their impact on the occurrence of HPAI-H5N1 outbreaks is comparatively less noticeable.

The findings of this study highlight the noticeable impact of species and economic trade on the occurrence of HPAI-H5N1 within the global poultry population. These factors are likely key contributors to the spread and transmission of the disease across various regions. Conversely, while climate and geographic landscape factors have some influence, they appear to play a less critical role in determining the outbreak patterns of HPAI-H5N1 in poultry. It should be noted, however, that further research is needed to fully understand the intricate dynamics of the disease's occurrence.

Our study identified chicken and duck density, population density, isothermality, and road density as the five most influential variables affecting global poultry HPAI-H5N1 outbreaks (Supplementary Figure S3). Chickens and ducks, being the primary susceptible species to HPAI-H5N1, transmit the virus not only among themselves but also to other mammals and humans. The response curve indicates that the development of large-scale, intensive

farming or farming communities correlates with a heightened risk of outbreaks. Interestingly, chicken density seems to have a more important influence on the occurrence of HPAI-H5N1 compared to duck density. This could be due to the more intensive breeding practices associated with chicken farming, as opposed to the more common backyard or free-range settings for duck breeding, which typically involve lower levels of intensification (Chowdhury et al., 2019). The higher chicken density in commercial farming systems may increase the likelihood of disease transmission and outbreak occurrences (Vijaykrishna et al., 2008).

Population density has emerged as a crucial factor in the global HPAI-H5N1 outbreak. There is a positive correlation between population density and outbreak risk, indicating that areas with higher population densities are associated with a greater risk of outbreaks. This correlation can be attributed to the increased demand and breeding volume for poultry in densely populated areas, as poultry is a major source of meat for the world's population. Population density encapsulates various elements, including human activities, live poultry trade, and environmental climate, all of which contribute to the outbreak risk. Additionally, secondary transmission of AIV is often linked to human involvement (Alexander, 2007), with human commercial activities playing an important role in the global spread of HPAI-H5N1 (Sealy et al., 2019). Commercial activities related to poultry are major determinants of its global dispersion (Gauthier-Clerc et al., 2007).

The study's analysis also highlights the impact of isothermality on the risk of HPAI-H5N1. The corresponding curves clearly show that the risk of HPAI-H5N1 increases with higher isothermality,

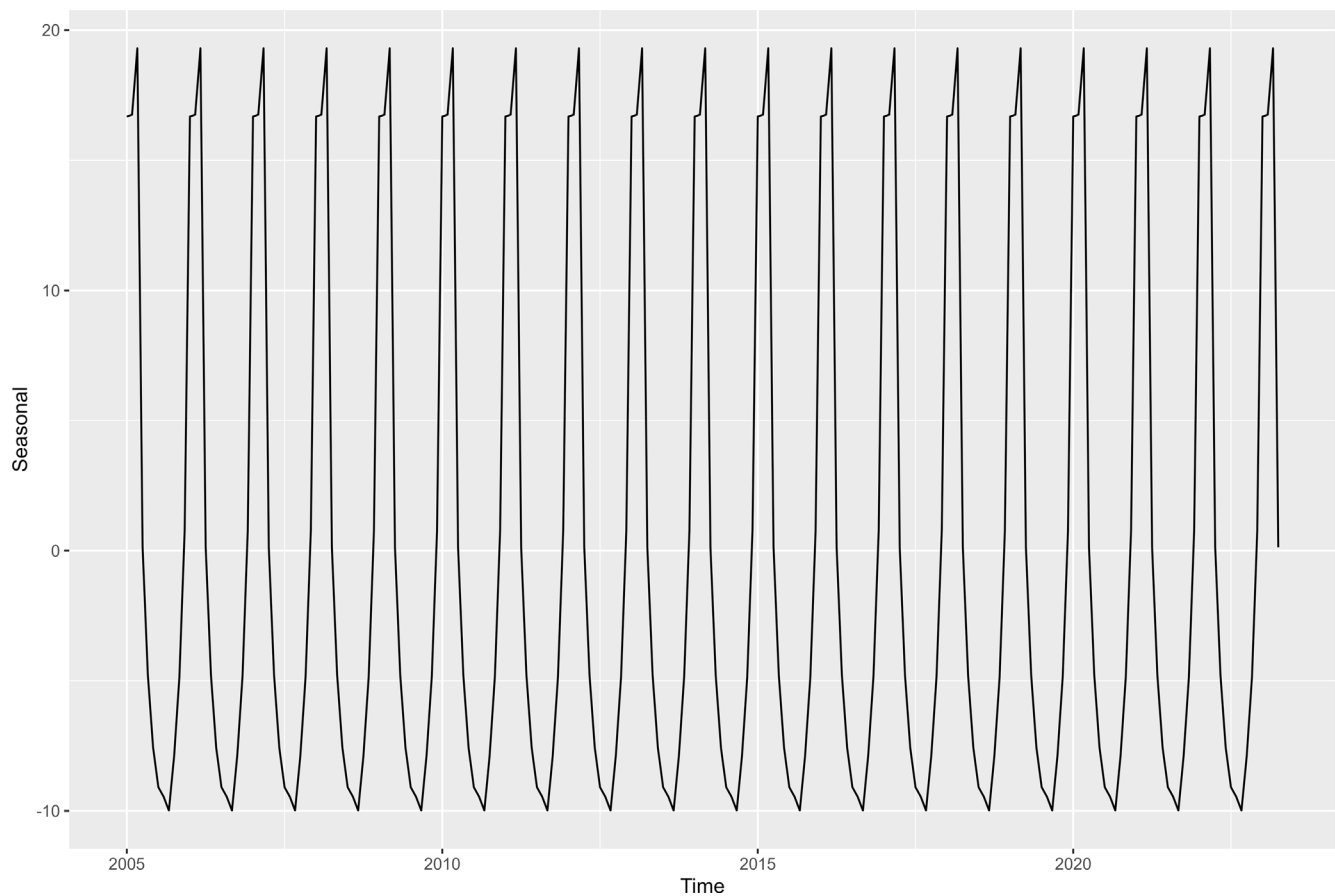


Fig. 3. Seasonal components of the event sequence of monthly incidence events reported for HPAI-H5N1 in global poultry from 2005 to 2023.

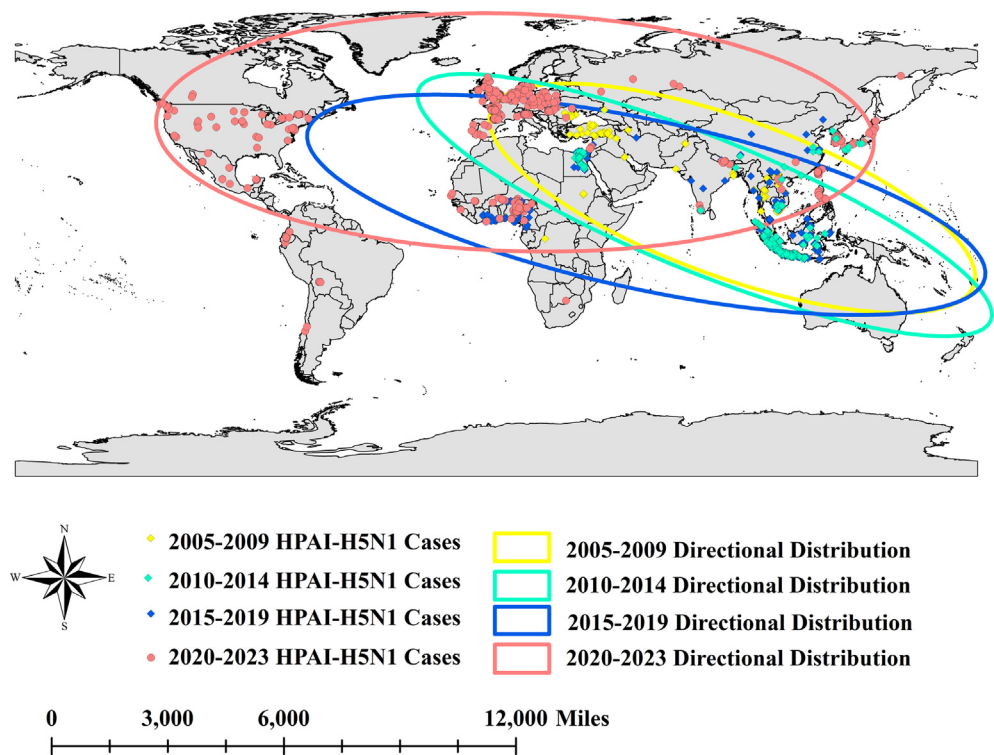


Fig. 4. Analysis of the directional distribution of HPAI-H5N1 disease in global poultry from 2005 to 2023.

Table 1
Parameters related to the directional distribution analysis for each stage of HPAI-H5N1 (highly pathogenic avian influenza virus), using SD ellipses.

Phase	Center coordinates	X- axis distance (km)	Y- axis distance (km)	Rotation angle (°)
Phase 1 2005–2009	(88.20E, 13.94 N)	96.06	29.55	109.87
Phase 2 2010–2014	(77.84E, 11.36 N)	115.46	25.11	111.95
Phase 3 2015–2019	(56.01E, 11.45 N)	129.66	31.60	102.15
Phase 4 2020–2023	(6.89E, 38.49 N)	134.34	44.33	91.82

Table 2
The variables included in the ecological niche model and their contribution rate of HPAI-H5N1 (highly pathogenic avian influenza virus), using maximum entropy.

Variable code	Variable Name	Variable contribution (%)
Chicken	Chicken density	47.6
Duck	Duck density	18.5
People	People density	16.9
Bio_3	Isothermality (BIO2/BIO7) (×100)	4.7
Road	Road density	2.3
Bio_19	Mean Precipitation of Coldest Quarter	1.8
Land cover	Land cover	1.7
Railways	Railways density	1.3
Bio_14	Mean Precipitation of Driest Month(mm)	1.1
Bio_15	Precipitation Seasonality	1
Elev	Elevation/Altitude (m)	0.9
Bio_10	Mean Temperature of Warmest Quarter (° C)	0.9
Bio_9	Mean Temperature of Driest Quarter (° C)	0.8
Bio_2	Mean Diurnal Range (° C)	0.3
Bio_18	Mean Precipitation of Warmest Quarter	0.1

peaking at an 83% level of isothermality. Isothermality is an index that quantifies the extent of annual oscillation of diurnal temperature relative to the temperature extremes of the hottest and coldest months. It represents the ratio of mean diurnal range to the temperature annual range. A larger ratio indicates more consistent annual temperatures with less pronounced seasonal temperature variations.

Road density is a noticeable indicator of trade, as areas with higher road density typically experience increased traffic flow of people and vehicles, leading to heightened trade activity and often correlating with larger populations. The response curve from our

study revealed a continuous increase in the risk of HPAI-H5N1 outbreaks with rising road density. This correlation aligns with findings from previous studies, which have identified road density or proximity to roads as critical factors in HPAI-H5N1 outbreaks (Ward et al., 2008; Shih et al., 2022). Roads often serve as transport networks for live poultry, potentially facilitating the spread of AIV, particularly along major national routes (Yang et al., 2020).

Our study conducted a comprehensive mapping of high-risk areas for HPAI-H5N1 outbreaks in poultry on a global scale. The results highlight West Africa, Europe, Southeast Asia, and southern China as regions with a high risk of HPAI-H5N1 outbreaks.

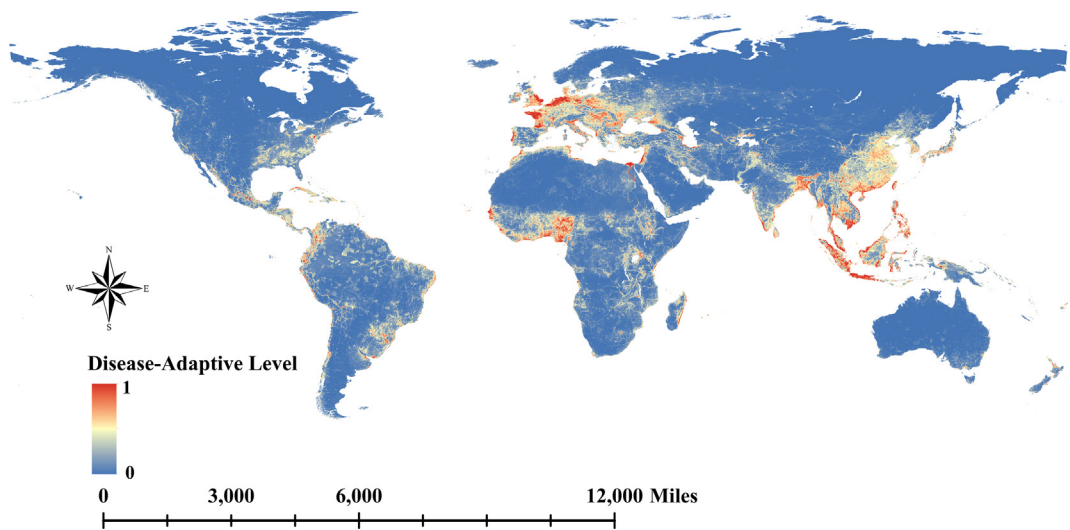


Fig. 5. Use ecological niche model to derive the risk level of HPAI-H5N1 (highly pathogenic avian influenza virus) outbreak in the globe. Note: 0 represents no risk and 1 represents the highest risk.

Increased vigilance and proactive measures are crucial in these areas to prevent and control the spread of the disease. Additionally, while the Americas, northern China, and eastern China currently exhibit relatively lower risk levels, recent case reports suggest a growing number of HPAI-H5N1 occurrences in these regions (Kandeil et al., 2023; Ke et al., 2022; Yang et al., 2023). It is, therefore, important to maintain vigilance and implement effective measures in these areas as well to mitigate the risk of HPAI-H5N1 outbreaks.

In 2005, a discovery was made at Qinghai Lake, a crucial breeding ground for wild birds along the Central Asian migration route: wild migratory birds killed by AIV were found (Li et al., 2017). However, not all migratory birds succumb to the HPAI virus; those that survive can potentially spread AIV to other areas along their migratory (Lycett et al., 2019). Between 2005 and 2008, numerous cases of H5N1-HPAI in wild birds were reported across Europe (Si et al., 2021), underscoring the important role of wild birds in the dissemination of AIV (Grillo et al., 2015).

Despite this, the complexity and lack of precise distribution locations of wild birds' migration routes posed important challenges in incorporating this factor into our study model. Consequently, we opted to exclude wild bird species as a variable in this study. Nevertheless, it is important to acknowledge that the migration of wild birds remains a critical factor in the spread of avian influenza.

This study does have some limitations. While the data were obtained from official sources, the variability in monitoring and reporting standards across different countries could introduce bias into the model's results. We considered several variables, such as wind speed, elevation, and land cover, which are pertinent to the ecology of wild birds but can only partially reflect their habits. Additionally, while the maximum entropy ecological niche model effectively represents the distribution of viral fitness, it is important to note that the global scope of this study means the results might not be universally applicable to every country, considering the differences in policies and poultry breeding patterns.

Conclusion

This study utilized global data on poultry HPAI-H5N1 outbreaks from January 2005 to April 2023 to conduct an in-depth analysis, including time-series decomposition and directional distribution assessment. We integrated a range of factors — species variables, economic and trade factors, bioclimatic factors, and geographical landscape variables — to predict high-risk areas and identify key risk factors for global poultry HPAI-H5N1 outbreaks. Our findings revealed that HPAI-H5N1 outbreaks exhibit both cyclical and seasonal patterns, with a predominant northwest-southeast direction in their spread. Critical factors influencing the occurrence of the disease were identified as chicken density, duck density, population density, isothermality, and road density. The model developed in this study effectively pinpoints high-risk areas for poultry HPAI-H5N1 outbreaks, thereby providing a valuable foundation for the prevention and control of disease outbreaks.

Supplementary material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.animal.2024.101085>.

Ethics approval

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered

to. No ethical approval was required, as this article does not include any research on live animals or biological samples.

Data and model availability statement

The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) did not use any AI and AI-assisted technologies.

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Declaration of interest

The authors declare no conflict of interest with regard to this study.

Acknowledgments

Research reported in this study was supported by Heilongjiang Provincial Key Laboratory of Pathogenic Mechanism for Animal Disease.

Financial support statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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